ECON 101: Macroeconomics
Solutions – Quiz 4

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Multiple Choice (1 mark each)

1. Rule of 70

Correct: A. The rule of 70 is a shorthand for doubling time under constant growth:

$$\text{Years to double} \approx \frac{70}{g} \quad (g \text{ in percent}).$$

Why: Starting from Y_0, with constant percentage growth g, $Y_t = Y_0(1+g)^t$. "Doubling" means $Y_t = 2Y_0$, so $(1+g)^t = 2$. Taking natural logs gives $t = \ln 2/\ln(1+g) \approx 0.693/g$, and $0.693 \times 100 \approx 69.3 \approx 70$ when g is in percent.

- B is false (confuses a heuristic with an actual annual growth rate).
- C refers to the price level, not output; unrelated to the rule.
- D confuses productivity with input scaling; not the rule's content.

2. Long-run labour productivity

Correct: B. Training and education raise *human capital*, shifting the per-worker production function upward and increasing output per hour in the long run.

- A (temporary tax rebate) mainly affects short-run consumption, not long-run productivity.
- C (higher nominal wage) is a price of labour; by itself it doesn't raise productivity.
- D (higher money supply) is a nominal change; long-run productivity is real and driven by technology, human/physical capital, and institutions.

3. Higher national saving (policy) and long-run effects

Correct: B. More national saving increases the supply of loanable funds, lowers the real interest rate, and *raises investment*. Higher capital deepening (and often technology adoption) increases the growth rate of potential output.

- A is the opposite of the standard loanable-funds channel.
- C is false: the capital stock and potential output shift upward as investment accumulates.
- D is ambiguous and typically incorrect in the long run: higher saving implies lower current consumption and higher investment.

4. When planned AE < actual Y

Correct: C. If planned spending is below current output, unsold goods accumulate as *unplanned inventory investment*. Firms respond by *cutting production*.

- A is the opposite of what firms do when inventories rise unexpectedly.
- B is wrong: inventories *increase*, not decrease.
- D (raising prices) does not resolve an excess supply of goods in this simple AE framework.

5. What lowers planned investment?

Correct: C. A deterioration in business confidence reduces expected future profitability, lowering planned investment even at a given real interest rate.

- A: a decline in the real interest rate *raises* investment (cheaper financing).
- B: higher expected profitability *raises* investment.
- D: technology improvements typically *raise* investment (new profitable projects).

Problem 1 (Growth and doubling time)

Given: Real GDP per capita = \$50,000 initially. Consider two constant annual growth rates.

(a) Doubling time at 2% per year. We use both the exact formula and the rule of 70 for clarity.

$$(1+g)^t = 2 \quad \Rightarrow \quad t = \frac{\ln 2}{\ln(1+g)}.$$

With $g = 0.02$,

$$t = \frac{\ln 2}{\ln(1.02)} \approx \frac{0.6931}{0.01980} \approx 35.0 \text{ years}.$$

Rule of 70: $70/2 = 35$ years, confirming the exact calculation.

(b) Doubling time at 3% per year.

$$t = \frac{\ln 2}{\ln(1.03)} \approx \frac{0.6931}{0.02956} \approx 23.5 \text{ years}.$$

Rule of 70: $70/3 \approx 23.3$ years (very close).

(c) Why small growth-rate differences matter. Because growth compounds, a 1 percentage point increase in g cuts the doubling time by roughly a third (from ~ 35 to ~ 23.5 years). Over generations, the level of income per person diverges dramatically: with $g = 2\%$, after 70 years,

$$\frac{Y_{70}}{Y_0} = (1.02)^{70} \approx 4.0,$$

whereas with $g = 3\%$,

$$\frac{Y_{70}}{Y_0} = (1.03)^{70} \approx 7.6.$$

Thus, small changes in g produce large differences in living standards over time.

Problem 2 (Aggregate Expenditure equilibrium without using the multiplier)

Given components (billions):

$$\begin{aligned} &C = 250 + 0.8Y_d, \quad Y_d = Y - T, \\ &I = 150, \quad G = 200, \quad NX = -50, \quad T = 100. \end{aligned}$$

(a) Planned AE as a function of Y. By definition,

$$AE = C + I + G + NX.$$

Substitute C and Y_d:

$$\begin{aligned} AE &= [250 + 0.8(Y - T)] + 150 + 200 + (-50) \\ &= 250 + 0.8(Y - 100) + 150 + 200 - 50 \\ &= 250 + 0.8Y - 80 + 150 + 200 - 50 \\ &= (250 - 80 + 150 + 200 - 50) + 0.8Y \\ &= 470 + 0.8Y. \end{aligned}$$

(If you obtained $440 + 0.8Y$*, you subtracted* 80 *twice; the correct constant term is* 470*.)*

(b) Equilibrium output Y where $AE = Y$. Set $Y = AE(Y)$:

$$Y = 470 + 0.8Y \quad \Rightarrow \quad Y - 0.8Y = 470 \quad \Rightarrow \quad 0.2Y = 470$$
$$\Rightarrow \quad Y = \frac{470}{0.2} = 2350.$$

(c) Household saving at equilibrium. First compute disposable income and consumption:

$$Y_d = Y - T = 2350 - 100 = 2250,$$
$$C = 250 + 0.8Y_d = 250 + 0.8(2250) = 250 + 1800 = 2050.$$

Household saving S is disposable income minus consumption:

$$S = Y_d - C = 2250 - 2050 = 200.$$

(d) Government budget balance at equilibrium. Budget balance $= T - G = 100 - 200 = -100$ (a deficit of \$100 billion).

Consistency checks:

- National income identity in an open economy:

$$Y = C + I + G + NX = 2050 + 150 + 200 - 50 = 2350 \quad \checkmark$$

- Leakages = injections:

$$\underbrace{S}_{200} + \underbrace{T}_{100} + \underbrace{(-NX)}_{50} = 200 + 100 + 50 = 350$$
$$\underbrace{I}_{150} + \underbrace{G}_{200} = 350 \quad \checkmark$$

  Both conditions hold at the computed equilibrium $Y = 2350$.

Instructor note: The equilibrium constant term is 470, yielding $Y = 2350$. This aligns all sectoral balances and avoids the arithmetic slip that leads to $Y = 2200$.